Capacity Eviction in Reconstruction-Trained Tokenizers: A Controlled Study on Financial Microstructure

Lakshay Bhati
[affiliation]
[email]

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Abstract

Two-stage models for financial time series learn a tokenizer under a reconstruction objective and then fit an autoregressive model to the resulting discrete codes. This design carries an unstated assumption: that a representation good enough to reconstruct from is good enough to forecast from. We show that it fails, and that it fails in a specific, predictable direction. Reconstruction objectives allocate code capacity by variance and covariance and never by downstream value, so wherever the code budget is tight enough that not every dimension can be coded and the competing dimensions are correlated enough to amortize one another, a low-variance channel that is weakly covariant with price — which is what microstructure is — is evicted deterministically rather than by chance. On an entropy-calibrated fixture in which every non-price dimension carries identical marginal variance and differs only in cross-dimensional dependence, per-dimension reconstruction sorts by variance class: 0.98 for the high-variance return channel, 0.82–0.92 for the mutually correlated block, and 0.001–0.014 for the one independent dimension, with the ordering near-identical across three seeds. A signal worth 1.151 nats planted in feature space is not recovered at all (probe correlation below 0.05 at all 40 evaluations), while 0.900 nats planted directly in token space is recovered by the identical backbone at 94.4% — localizing the loss to the tokenizer–backbone interface rather than to model capacity. We re-specify that interface so the fine subtoken is a per-bar encoding of the current bar, restoring per-bar legibility from 0.5135, against a chance rate of 0.50, to 0.9047 under a gate pinned at 0.90, and recovering the original feature-space signal at unchanged bits per token. We then evaluate whether real trade-flow imbalance survives this pipeline in a pre-registered, capacity-matched five-cell ablation with a shuffled-microstructure placebo, on 304,625,181 one-minute bars across 200 instruments, scored by cost-aware net information ratio: **[verdict]**. The design's minimum detectable effect is 3.518 annualized IR against a 0.5 economic floor, and the only prior we hold for the quantity under test is our own univariate screen of trade-flow imbalance on a single symbol, $|\text{RankIC}| = 0.027$ at $h = 5$ and 0.020 at the headline horizon — so a null or inconclusive outcome is the most likely honest one. Both were pre-committed as publishable before any result existed. Microstructure-awareness is an architectural property of a tokenizer, not an emergent one.

1 Introduction

Contributions.

- We identify and mechanistically characterize *capacity eviction*: under a reconstruction objective, code capacity is allocated by variance and covariance and never by downstream value, so a task-relevant channel that is low-variance and weakly covariant is excluded deterministically. We isolate the effect with a fixture in which the evicted dimension and the retained ones differ in exactly one property — statistical independence — and show the exclusion reproduces across seeds (Section 3).
- We localize the failure by construction rather than by inference: the same backbone at the same budget recovers 94.4% of a signal planted in token space and none of a larger signal planted in feature space, which acquits model capacity and indicts the tokenizer–backbone interface (Section 3.4).
- We introduce a per-bar tokenizer interface — a pointwise fine subtoken with a bottleneck decode leg and calibrated class weighting — that restores per-bar legibility at unchanged bits per token, and a *legibility gate* that enforces the property on real data as a pre-second-stage hard stop (Sections 3.6 and 3.7).

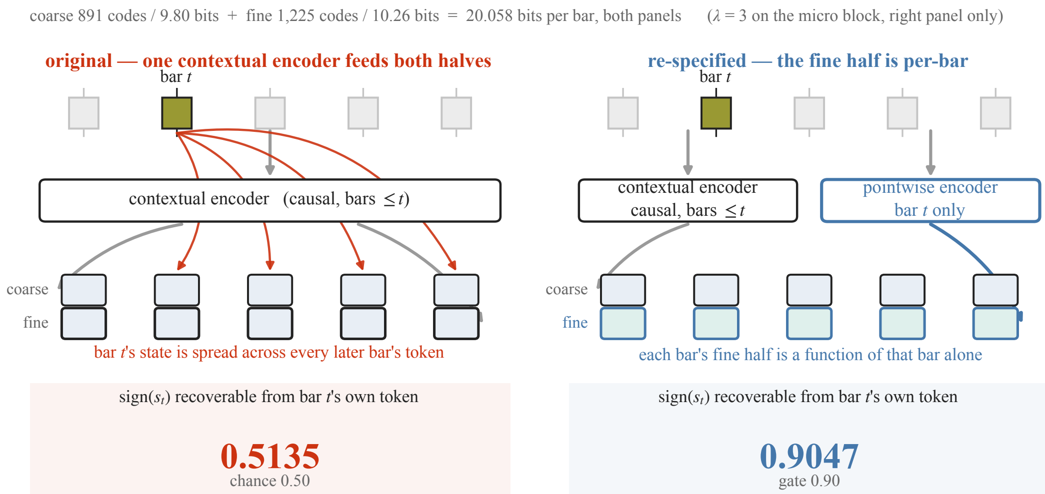

Figure 1: One interface change decides whether a microstructure signal survives tokenization. Each bar is encoded to a token pair — a coarse and a fine subtoken, together 20.058 bits — which is what the autoregressive stage consumes. **Left, the original design:** a single causal encoder produces both halves. Because that encoder is contextual, bar t's own state is distributed across the tokens of every later bar in the window, so the window reconstructs while the per-bar symbol carries almost nothing: the sign of bar t's microstructure state is recoverable from bar t's own token at 0.5135, against a chance rate of 0.50. **Right, the re-specified design:** the fine half is produced by a pointwise encoder that sees only the current bar, shaped by a per-bar bottleneck objective that weights the microstructure block as a class. Recoverability rises to 0.9047 and is enforced thereafter by a gate at 0.90. Bits per bar, the vocabulary split, the backbone and every evaluation instrument are identical between the two panels, so per-bar legibility is the only quantity that differs. What that difference buys downstream — the same planted signal recovered on the right and not on the left — is Figure 3; this panel is the interface, not the result.

- We run a pre-registered, capacity-matched five-cell ablation with a shuffled microstructure placebo on 304,625,181 one-minute bars, scored by cost-aware net information ratio under purged walk-forward validation, with the decision rule, seed count and multiple-testing correction fixed before any result existed (Sections 4 and 6).

2 Related work

[§2 — four clusters: K-line foundation models; discrete tokenization (VQ/FSQ/BSQ); microstructure prediction with the TFI-vs-OFI distinction; evaluation discipline. Drafted after §4, §5, §7, §8.]

3 Capacity eviction in reconstruction-trained tokenizers

Two-stage models for financial time series learn a tokenizer under a reconstruction objective and then fit an autoregressive model to the resulting discrete codes [Shi et al., 2025, van den Oord et al., 2017]. The design carries an assumption that is seldom stated: that *a representation good enough to reconstruct from is good enough to forecast from*. The tokenizer is fitted to reconstruction, the forecaster never sees the underlying series, and nothing in the pipeline is trained to preserve what the forecaster will need.

We show that the assumption fails, that it fails deterministically rather than by chance, and that it fails hardest on precisely the channels a microstructure-aware model depends on. We state the mechanism (Section 3.1), measure it (Section 3.2), test the prediction it makes about an end-to-end pipeline (Section 3.3), localize the loss by construction rather than by inference (Sections 3.4 and 3.5), give an intervention that removes it (Section 3.6), and convert the result into a standing gate that must pass on real data before any second-stage expenditure (Section 3.7). Every measurement in this section is made on a synthetic fixture with a known ground-truth signal, chosen so that the quantity being recovered is exactly computable; Section 3.8 states what that does and does not license.

3.1 The mechanism

A tokenizer maps each bar to a code of fixed size, so capacity is scarce and must be allocated across the bar's dimensions. The allocation is chosen by whatever the objective rewards, and a reconstruction objective rewards two things.

> **Claim.** Under a reconstruction objective, code capacity is allocated in order of the loss reduction each dimension buys. That ordering is set by a dimension's *variance*, which bounds how much loss it can contribute, and by its *covariance* with dimensions already coded, which determines how cheaply it can be served by capacity already spent. Downstream predictive value enters nowhere. A dimension that is low-variance *and* weakly covariant with the rest is therefore the last claim on the budget. Where two conditions hold together — a budget tight enough that not every dimension can be coded, and competing dimensions correlated enough that coding them is partly amortized — that dimension receives nothing, and does so deterministically: not occasionally, but in every run. Where either condition fails, the same objective still allocates by the same rule, but the exclusion is no longer determined; Section 3.2 reports a fixture in which the second condition is absent and the allocation is an initialization lottery instead.

The budget is tight, and the arithmetic says by how much. The canonical configuration uses per-dimension levels $(11, 9, 9, 7, 7, 5, 5)$, split into a coarse subtoken of $11 \times 9 \times 9 = 891$ codes (9.80 bits) and a fine subtoken of $7 \times 7 \times 5 \times 5 = 1225$ codes (10.26 bits), for 20.058 bits per bar; the binary-spherical arm [Zhao et al., 2024] spends $10 + 10$ bits for 20.000. Recovering the sign of a jointly-normal dimension at accuracy a requires a correlation $\rho = \cos\big(\pi(1 - a)\big)$, so the $a = 0.90$ we will require costs at least $-\frac{1}{2}\log_2(1 - \rho^2) = 1.69$ bits for that dimension alone by the Gaussian rate–distortion bound, and more under any realizable scalar quantizer. Thirteen live dimensions at that fidelity would need at least 22.0 bits — more than the entire per-bar budget, at any coarse/fine split. Not every dimension can be coded well. Something must be dropped, and the objective decides what.

This matters here because microstructure has exactly the profile the claim identifies. Trade-flow imbalance and the aggregate trade-flow statistics are low-variance relative to returns and only weakly covariant with price shape — they carry information about the near future precisely because they are *not* a restatement of the recent past. A tokenizer trained to reconstruct will therefore allocate capacity away from them by construction. This is not a defect of the objective; it is the objective working correctly against a goal it was never given. It also means that "microstructure-aware" has to be an architectural property of a tokenizer, and cannot be an emergent one.

3.2 Measuring the eviction

To measure the claim we need a fixture in which one dimension differs from the others in exactly the property the claim names, and in nothing else. We construct a synthetic bar stream in which every dimension is marginally standard normal — so all of them contribute equally to a variance-weighted loss — and then make eleven of them mutually dependent while one, the *signal* dimension, is independent of everything.

The dependence is not chosen for convenience; it is calibrated to the real data. On 30 symbols and 3,895,808 unmasked bars, the Gaussian-copula correlation log-determinant over the seven non-microstructure dimensions is -6.4819, a per-dimension entropy deficit of 0.4631 nats. The fixture reproduces that deficit with a cross-dimension AR(1) chain at $\rho = 0.7993$, achieving 0.462 nats against a 0.05-nat tolerance. One further dimension, the return, is driven by the signal dimension at a two-bar lag and so carries $3.15\times$ the marginal standard deviation of the rest: it is the high-variance channel, and it is also the only dimension with downstream value in this fixture, since it is what a forecaster would predict.

Figure 2 and Table 1 report, for three independently initialized tokenizers, how well each dimension can be recovered from the bar's own code. The result sorts by variance class and does not vary

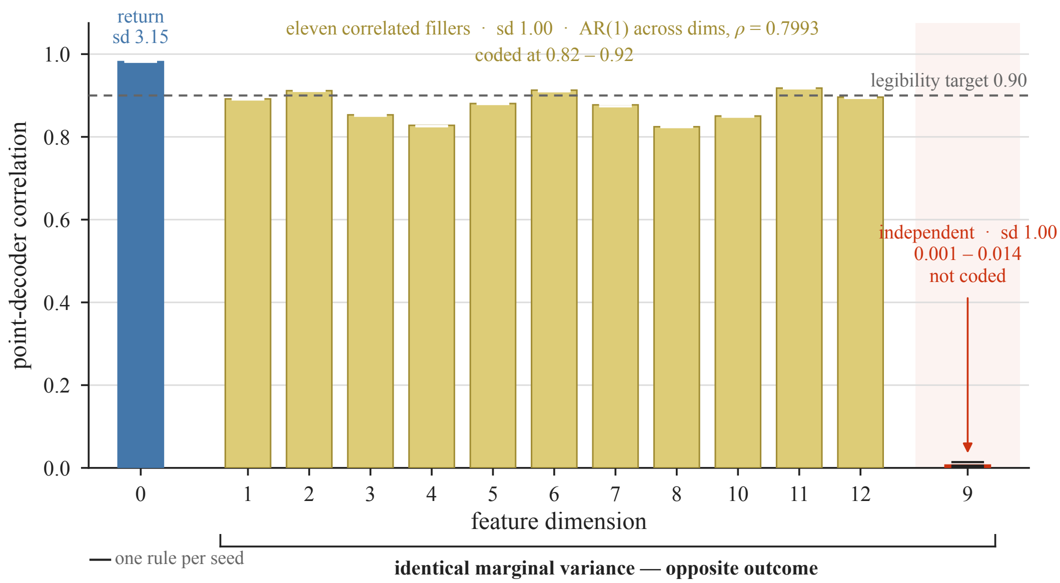

Figure 2: Reconstruction quality is sorted by variance class, not by task relevance, and the ordering is deterministic. Point-decoder correlation between each dimension's value at bar t and its reconstruction from bar t's own code, for three independently initialized tokenizers on a fixture whose non-signal structure is entropy-calibrated to the real data. The return dimension, whose marginal standard deviation is 3.15× that of the others, is recovered at 0.98; four of the thirteen dimensions clear the 0.90 rule in every seed. The eleven filler dimensions, which have unit marginal variance and are coupled by an AR(1) chain across dimensions at $\rho = 0.7993$, are recovered at 0.82–0.92. The signal dimension has *the same unit marginal variance as the fillers* and differs from them in exactly one property — it is statistically independent of everything — and is recovered at 0.001–0.014. The three seeds agree to within the marker width, so this is deterministic exclusion rather than an initialization lottery. Reconstruction objectives buy variance and covariance; they do not buy independence.

Table 1: Reconstruction quality by variance class, three seeds. Point-decoder correlation between a dimension's true value at bar t and its reconstruction from bar t's own fine code. Every dimension except the return has unit marginal variance; the signal dimension differs from the fillers only in being statistically independent. Filler entries give the range over the eleven filler dimensions; per-dimension values are in Appendix A.

dimension class	cross-dimensional structure	seed 0	seed 1	seed 2
return, sd 3.15	driven by the signal at lag 2	0.9818	0.9815	0.9823
fillers (11), sd 1.00	AR(1) chain across dims, ρ = 0.7993	0.826–0.918	0.824–0.919	0.824–0.917
signal, sd 1.00	independent	**0.0140**	**0.0011**	**0.0053**

across seeds. The return is recovered at 0.98. The eleven mutually dependent dimensions are recovered at 0.82–0.92, because one shared component serves all eleven at once and the capacity spent on it is amortized eleven ways. The signal dimension — same marginal variance as any single filler, differing only in being independent — is recovered at 0.001–0.014. It is not coded at all.

Calibrating the fixture to the real data is what makes this a mechanism rather than an anecdote, and we report the correction it forced because it changes the strength of the claim. On an earlier fixture whose dimensions were exchangeable i.i.d. draws — no cross-dimensional dependence at all — the allocation was arbitrary: across three initializations the set of dimensions clearing 0.9 changed completely, and the dimension of interest cleared in one seed of three. That reading makes the phenomenon an initialization lottery, which is a substantially weaker and less actionable claim. Introducing the dependence structure the real data actually has replaced the lottery with deterministic exclusion, because the correlated block then offers an amortization the independent dimension cannot match.

3.3 The prediction, tested

If a tokenizer evicts a channel then a signal carried only by that channel is unavailable to everything downstream, however much information it contains and however capable the downstream model is. We test that prediction by planting a signal of known size.

We build a three-symbol stream of 42,000,000 bars. The signal dimension s_t is drawn i.i.d. standard normal and observed at the close of bar t; in the *planted* arm it drives the return two bars later,

$$r_t \;=\; \sigma\big(\varepsilon_t + c\, s_{t-2}\big), \qquad c = 3, \quad \varepsilon_t \sim \mathcal{N}(0,1) \text{ i.i.d.}, \tag{1}$$

and in the *noise* arm s_t is redrawn independently, so the arms differ in exactly one respect. Because the states are i.i.d., past returns reveal only states at lag ≥ 2, which are orthogonal to the forward label: the plant is recoverable *only* through the microstructure channel, and not through any function of price history. Equation (1) is a Gaussian channel, so the information it injects is exact,

$$I(s_t;\, r_{t+2}) \;=\; \tfrac{1}{2}\ln\Big(1 + c^2\Big) \;=\; \tfrac{1}{2}\ln 10 \;=\; 1.151 \text{ nats per bar}, \tag{2}$$

and because the tokens are a deterministic function of the features, the data-processing inequality makes 1.151 nats an upper bound on the validation cross-entropy improvement the autoregressive stage can obtain from the plant.

Both arms were trained under an identical budget and schedule — 20,000 steps at batch 32 and context 32, giving 20,480,000 bar-visits against a 27,300,000-bar training region, so 0.75 epochs, and memorization is not available as an explanation. The sub-epoch condition is asserted in the run receipt rather than assumed.

The pipeline extracted none of it. The schedule completed on both arms with no loss spikes; the validation–train gap was +0.0072 (noise) and −0.0052 (planted), confirming the sub-epoch regime; the correlation between the model's forecast and the planted state stayed below 0.05 at *every one of the* 40 *evaluation points*, with a maximum of 0.0433; and the planted arm's final validation NLL was 13.6113 against the noise arm's 13.5979 — worse by 0.0134 nats, which is the wrong sign.

The tokenizer was not discarding the state. Window-level reconstruction of the planted dimension was non-degenerate in the same run: MAE 0.3044 against a predict-the-mean baseline of 0.8216. The information was present in the code stream and unavailable to the model consuming it.

3.4 Localizing the loss: a token-space control

A null admits two explanations: the tokenizer's code geometry hides the rule, or the backbone cannot learn thin conditionals at this scale. We separate them by construction rather than by argument, planting a signal of comparable size *directly in token space* — bypassing the tokenizer entirely — and asking whether the same backbone at the same budget recovers it.

The plant redraws a target digit from a discretized Gaussian centred monotonically on a source digit, with its width solved so the mutual information lands in a pre-registered band. It delivers 0.9003 nats exact over the full stream, agreeing with a Monte-Carlo plug-in estimate to 2.0×10^{-5}, with a marginal KL of 0.0048 and an entropy shift of −0.005 nats — so no information is smuggled in through the marginal — and an unplanted control returns -8.8×10^{-5}. Construction, calibration, and the retirement of a first, information-poor plant design are given in Appendix B. The carrier is the regenerated noise stream of Section 3.3 tokenized by that run's frozen tokenizer, so token distribution, budget, schedule and validation protocol are unchanged, and that run's noise arm supplies the reference $H_0 = 13.5979$.

The backbone recovers it almost completely: the probe crosses the pre-registered detection threshold of 0.30 at step 2,000 and reaches 0.9999, and the final validation NLL is 12.7483, that is $H_0 -$ 0.8496 nats, or 94.4% of the planted information. Figure 3 places the two experiments side by side against their planted-information references. The backbone is not the limiting factor. The loss is at the tokenizer–backbone interface.

3.5 The interface defect, measured directly

The window encoder is causal but *contextual*: bar t's code is a function of bars $\leq t$. State belonging to bar t is therefore free to be distributed across the codes of every subsequent bar in the window.

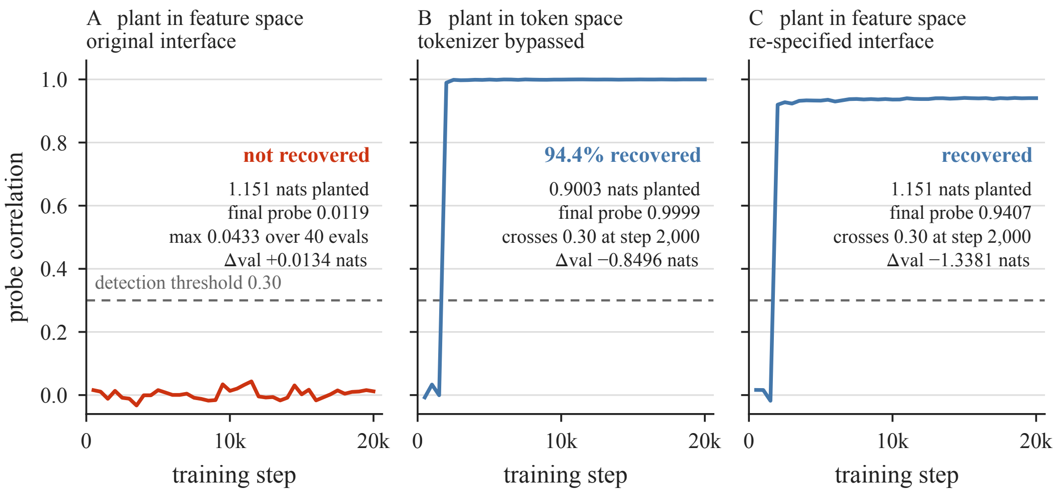

Figure 3: The same backbone extracts nearly everything from token space and nothing from feature space. Nats of validation cross-entropy recovered against training step, measured against a matched reference run, with the dashed rule in each panel marking the information actually planted. **Left:** 1.151 nats planted in feature space, where it must survive tokenization — the curve never leaves zero, and the forecast–state probe stays below 0.05 at all 40 evaluation points. **Centre:** 0.9003 nats planted directly in token space, bypassing the tokenizer — the identical backbone at the identical budget recovers 0.8496 nats, 94.4% of what was planted, crossing the detection threshold at step 2,000. **Right:** the original feature-space plant, after the tokenizer interface is re-specified (Section 3.6) — recovery returns. Model capacity was never the constraint; the interface was.

The window as a whole reconstructs; no individual code need carry its own bar's features. Since the autoregressive stage consumes exactly the per-bar code sequence, anything that lives only in that spread-out form is unreachable.

The measurement is unambiguous. On 300,000 bars of the planted stream, tokenized by the frozen tokenizer of Section 3.3 with a 240,000/60,000 split, a logistic probe recovering $\text{sign}(s_t)$ from bar t's own coarse and fine digits achieves 0.5135 against a chance rate of 0.5, and an MLP regressing s_t on the same digits achieves a Pearson correlation of 0.0142 — while the same tokenizer on the same stream reconstructs that dimension at the window level at MAE 0.3044 against a 0.8216 baseline. The state is encoded, and per-bar illegible.

A stronger form of the same defect appears when the encoder is not causal at all, and we record it because it is why causality is enforced. Perturbing only future bars inside a tokenization chunk and counting how many past bars' tokens flip gives a leakage rate of 41.75% coarse and 28.33% fine on a tokenizer trained on real data; the causal-encoder control measures exactly 0.0% on both, which is what establishes that the probe can read zero. Causal encoding is necessary for validity, and this section establishes that it is not sufficient for usability: it removes the leak forward and leaves the smear backward.

3.6 Intervention

If capacity is allocated by an objective that cannot see per-bar legibility, the remedy is to give the objective a term that can. We make three changes. Each was necessary; none was sufficient alone, and we report the intermediate failures because they are what establishes that.

(i) A pointwise fine encoder. The fine subtoken becomes a per-bar encoding of bar t's own features, produced by a branch with no cross-bar mixing, while the coarse subtoken retains the contextual causal encoder. Both properties are carried in one token pair; bits per bar, the vocabulary split and the cell design are untouched. Alone this changed nothing — legibility 0.5202, at chance. The optimizer routed the state through the contextual coarse smear and spent the fine bits elsewhere.

(ii) A per-bar bottleneck decode leg. A pointwise decode head must reconstruct bar t from bar t's fine code alone, with zero cross-bar access, and its loss is added to the training objective. This trains

— the dimensions that win reach point-decoder correlation above 0.9 — but it does not select *which* dimensions win: legibility 0.5136 and 0.5037 at canonical width, and 0.5014, 0.4995, 0.5182 across three seeds on the entropy-calibrated fixture, a 3/3 failure against the 0.90 threshold. That failure is the measurement plotted in Figure 2: the leg works, and the objective still spends its budget on variance and covariance.

(iii) Class weighting, with objective separation. The bottleneck loss weights the six microstructure dimensions as a class by a factor λ, and the window reconstruction losses receive the fine channels detached, so the fine encoder is shaped exclusively by the per-bar objective. Detachment is load-bearing, and was measured to be: with the window loss attached the per-bar weighting is inert at every λ — per-dimension receipts are identical from $\lambda = 2$ to $\lambda = 10^6$ — because the window objective re-creates the smearing incentive as fast as the leg removes it.

λ is calibrated, not chosen. Across 26 trials no (λ, β) pair cleared 0.90 on all three seeds, which places the threshold inside the instrument's seed-noise band at the achievable ceiling, while confirming that the state is carried (signal-dimension point-decoder correlation 0.77–0.96, microstructure class 0.86–0.92, return preserved above 0.96). The acceptance rule was therefore restated over the three calibration seeds as mean $\geq$ 0.90 and every seed $\geq$ 0.85, and λ set to the smallest searched value clearing it: $\lambda = 2$ gives $(0.8365, 0.8594, 0.8592)$, mean 0.8517 — fail; $\lambda = 3$ gives $(0.9060, 0.9142, 0.9000)$, mean 0.9067, minimum 0.9000 — pass. Re-running through the pinned construction path reproduces mean 0.9047, minimum 0.8839, bit-identically to the direct construction at fixed thread count.

We state plainly that this is an acceptance threshold restated after seeing calibration results, and why it is not the failure that description usually names. The restatement was forced by a measured property of the instrument rather than chosen to admit a preferred value: 0.90 sits inside the calibration instrument's own seed-noise band of roughly ± 0.03 at the achievable ceiling, so a fixed single-seed threshold at exactly that value tests the noise as much as the architecture, and the restated form — a mean clause plus a per-seed floor — is the standard response to that situation rather than a loosening of it. It occurred entirely on the synthetic calibration fixture, at a point when the real-data gate had never been executed and no real microstructure had been tokenized. And it did not propagate: the standing gate that governs expenditure (Section 3.7) requires $\geq$ 0.90 on *every one* of the six microstructure dimensions, with no mean clause and no floor, and nothing downstream of the calibration was selected on the restated form. $\lambda = 3.0$ is pinned in the conformance surface, applies identically to both quantizers and all input arms — so it cannot differentially favour any cell — and leaves bits per bar, the cell matrix and every evaluation instrument unchanged.

3.7 Verification, and the gate that carries it to real data

The intervention is verified by repeating Section 3.3 exactly: generator, seeds, stream size, plant strength, lag, budget, schedule and validation protocol unchanged, the tokenizer architecture the only difference. The forecast–state correlation crosses the detection threshold at step 2,000 and reaches 0.9407, with a maximum of 0.9414. The planted arm's final validation NLL is 11.8478 against the noise arm's 13.1859, a difference of -1.3381 nats where Section 3.3 gave $+0.0134$; the schedule completes without spikes and both arms stay in the sub-epoch regime. Per-bar legibility under the new tokenizer is 0.8990 on the planted arm and 0.9084 on the noise arm, where Section 3.5 measured 0.5135. We report Δval as the pre-registered decision statistic and not as an extraction fraction: each arm trains its own tokenizer, so the arm difference is not calibrated against the 1.151-nat plant in the way Section 3.4's control is calibrated against its own H_0. Figure 4 places every stage of the intervention against the gate.

A mechanism established on a fixture is a hypothesis about the real data, so we convert it into a gate rather than an expectation. After first-stage training and *before any second-stage expenditure*, each of the six microstructure dimensions must be linearly recoverable from bar t's own token pair at logistic sign-accuracy $\geq$ 0.90 on the run's real training stream; failure raises an exception and halts the run. The probe is a logistic regression on the one-hot digit decomposition of the coarse and fine codes, trained for 300 Adam steps at fixed seed on an 80/20 split, deterministic given its inputs. Sign accuracy is well defined for the magnitude-shaped dimensions because every microstructure feature enters z-scored, so the sign is above-versus-below the causal rolling mean, and per-dimension base

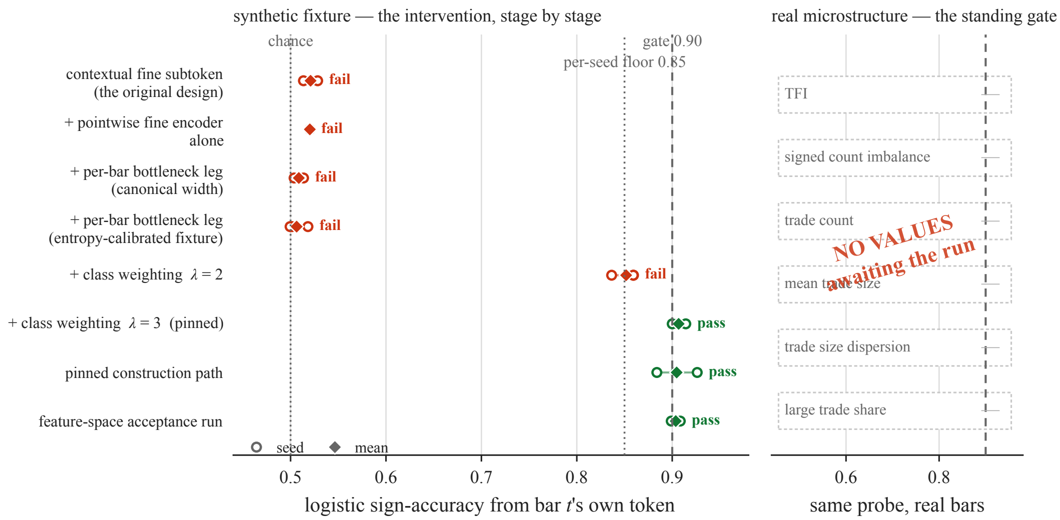

Figure 4: Only the third change moves per-bar legibility, and the threshold sits at the achievable ceiling. *Left:* logistic sign-accuracy of a microstructure dimension from bar t's own token, across the intervention. Open circles are seeds and the diamond is their mean; the pre-registered acceptance rule over the three calibration seeds is mean ≥ 0.90 and every seed ≥ 0.85, so both thresholds are drawn. The original contextual design, the pointwise encoder alone, and the pointwise encoder with the bottleneck leg all sit at chance — the leg trains, but does not decide which dimensions win. Class weighting at $\lambda = 2$ still fails; $\lambda = 3$ clears, and the pinned construction path reproduces it. Note how little headroom exists above the gate: this is the achievable ceiling, not a comfortable margin. *Right:* the same probe applied to real microstructure by the standing gate, which runs after first-stage training and before any second-stage expenditure. **[PH: no values drawn]** — the panel shows six empty slots rather than placeholder bars, because bars near the gate would imply a passing result nobody has measured. Six numbers drop in from the gate's own receipt, and the gate halts the run if any of them falls below the rule.

rates are recorded so that residual class imbalance is visible. The sample is stratified by symbol with the split blocked in time *within* each symbol, which preserves both properties that matter: full universe coverage, and no near-duplicate neighbours across the split. Two defects in this gate were found and closed before it was relied upon, both of the form "a check that cannot fail is not a check"; they are documented with their fixes in Appendix C.

The gate's first execution on real microstructure is a named checkpoint with a pre-written adjudication: if the real dimensions cannot clear the threshold under the calibrated architecture, the run stops and reports with per-dimension receipts. It is not a threshold that may be moved. We record the expectation before the fact. Real trade-flow imbalance has the low-variance, weakly-covariant profile that Figure 2 shows being evicted, and the correction was calibrated on a synthetic plant. **A gate failure on real data would be the real-data confirmation of the mechanism** — simultaneously a blocker for the ablation and the strongest available evidence for the claim of Section 3.1. Both outcomes are reportable, and neither would be a surprise.

3.8 Scope of the claim

Four limits, restated in Section 7. First, everything above is measured on a *synthetic fixture*, not on real microstructure; the fixture's non-signal structure is entropy-matched to the real data and its signal dimension is a controlled construction, which is what makes the recovered quantity exactly computable and is also what stops it being a measurement of markets. The mechanism is therefore not this paper's headline claim, and is not elevated to one under any experimental outcome. Second, we demonstrate the effect for one reconstruction objective, one quantizer family and one class of encoder: we show that *this* objective allocates by variance and covariance, not that no reconstruction objective can do otherwise — and Section 3.6 is itself a counterexample to the strong form, being a reconstruction objective that carries the evicted channel because it was given an explicit term to do so. Third, the 94.4% of Section 3.4 is calibrated against a single matched reference run and inherits whatever variation that run carries. Fourth, λ is a hyperparameter fitted to a synthetic proxy for the real allocation problem; it is pinned pre-data and applied identically across every arm, so it cannot favour any cell, but the gate of Section 3.7 exists precisely because the proxy might not transfer.

4 Experimental design

The mechanism of Section 3 says that a reconstruction-trained tokenizer will evict microstructure unless it is built not to. This section describes the experiment that asks the separate question the paper is organized around: whether real trade-flow imbalance, carried through a tokenizer built to preserve it, survives to a cost-aware trading decision. The design was frozen before any of its numbers existed, and this section describes it as frozen.

4.1 Five cells

The experiment is a 2×2 over the two factors that could plausibly explain a difference — the quantizer and the input arm — plus a fifth cell that isolates the one thing a 2×2 cannot: whether an effect attributed to microstructure is really microstructure. Figure 5 shows the layout; the registry that the runner reads is the authority.

Cells 1 and 2 receive the seven OHLC-shape, volume and amount dimensions only; cells 3 and 4 additionally receive the six microstructure dimensions (indices 7–12: trade-flow imbalance, signed count imbalance, trade count, mean trade size, trade-size dispersion, large-trade share). Cells 1 and 3 quantize with binary spherical quantization; cells 2 and 4 with finite scalar quantization. Cell 5 is cell 4 with the microstructure block shuffled, and is the placebo.

The comparison that carries the paper is the paired difference $\Delta\mathrm{IR}(4-5)$. The others are support: $\mathrm{IR}(4) - \mathrm{IR}(2)$ asks whether the microstructure arm beats the OHLCV arm without reference to the placebo, and the two BSQ cells anchor the quantizer axis. We are explicit in Section 7 that the three marginals involving a BSQ arm are withdrawn as claims, because the BSQ implementation has no external reference point; they are reported, not claimed.

In the OHLCV-only arm the six microstructure dimensions are *removed* from the input rather than supplied and left uninformative, so that arm's feature width is seven rather than sixteen. That choice matters for interpreting the legibility gate, which is undefined for an arm with no microstructure channel and is exempted there by a named registry entry rather than by silence.

4.2 Capacity is controlled, not assumed

A comparison between two quantizers is meaningless if one of them simply has more room. We therefore control the two quantities that determine room, and report both rather than asserting parity. The finite-scalar arm uses per-dimension levels $(11, 9, 9, 7, 7, 5, 5)$, split into a coarse subtoken of 891 codes and a fine subtoken of 1225, for 20.058 bits per bar; the binary-spherical arm spends $10 + 10$ bits for 20.000. The two differ by 0.058 bits per bar, inside a pre-registered parity band of 19.5–20.5.

The backbone realizes 21,301,248 parameters at the finite-scalar vocabulary and 21,231,616 at the binary-spherical one — a difference of 69,632 parameters, or 0.327%, arising entirely from vocabulary-dependent embedding and output-head rows. The backbone is otherwise identical across all five cells: eight layers, model width 512, feed-forward width 1024, eight heads, and a context of 512 bars. It is held fixed on purpose. The tokenizer is the object under study; the backbone is the instrument that measures it.

Every cell trains under the same budget: 26,003 steps at batch 32 and context 512, on the same draw. That budget is matched and fixed rather than chosen per cell by early stopping. The alternative — stopping each cell when its own validation loss saturates — would give five different budgets and confound $\Delta\mathrm{IR}(4-5)$ with "cell 4 trained longer", which is the same class of confound the placebo exists to remove. The cost is that no cell is guaranteed to be at its own optimum; saturation is measured and reported rather than optimized against.

4.3 The placebo, and the handicap it creates

The placebo answers a question no capacity control can: whether an effect attributed to microstructure comes from microstructure's *information* or merely from the model having six more input channels to spend capacity on. Cell 5 receives the same six channels, with the same marginal distributions, carrying no usable information.

The shuffle permutes the six microstructure dimensions as a block, within each contiguous segment of a symbol, with a full within-segment permutation rather than a fixed block length. Three properties are enforced rather than assumed. The block moves as a unit, so within-bar structure *among* the microstructure dimensions survives and only their relationship to everything else is destroyed. The mask travels with the value, because permuting values alone would strand fill values on valid bars and real values on masked ones — a defect found in review and fixed before any training. And the permutation is derived from the run seed and the symbol together, so the train and evaluation paths shuffle identically.

Figure 5 reports what the shuffle measurably does. It destroys the contemporaneous link between microstructure and price shape, from mean absolute correlation 0.2037 to 0.0005, and it destroys microstructure's own lag-1 autocorrelation, from 0.4165 to 0.0006. It leaves the within-block correlation at 0.2886 and the within-OHLCV correlation at 0.3105, both unchanged.

This is where the design's principal known weakness lives, and we state it rather than let a reader find it. Because cell 4's microstructure dimensions are correlated with price at 0.20 and autocorrelated at 0.42, part of the capacity they consume is *amortized* against structure the code is already carrying. Shuffling removes that amortization along with the information, so cell 5 pays a capacity price cell 4 does not. The consequence is a forfeit stated in the pre-registered rule string itself: $\Delta\mathrm{IR}(4-5)$ is (information) + (capacity handicap), and the paper does not claim the information effect alone clears any threshold. The handicap is not quantified in information-ratio units, and Section 7 carries that as an open disclosure rather than a resolved question.

4.4 The headline metric

The headline is a cost-aware net information ratio, annualized, computed on a portfolio return series rather than on pooled per-position returns. For a position of sign s opened at the close of bar $t+1$ and closed h bars later, the realized net log-return subtracts execution and funding cost directly:

$$r^{\mathrm{net}} = s \log \frac{C_{t+h}}{C_{t+1}} - c_{\mathrm{total}} - c_{\mathrm{funding}}, \qquad c_{\mathrm{total}} = 2\, c_{\mathrm{side}}. \tag{3}$$

Each period's portfolio return is the equal-weighted mean over the symbols active in that period, and the information ratio is computed on that series, so a symbol that is not trading contributes nothing rather than contributing a zero.

The per-side cost has three additive components — a taker fee, a volatility-scaled half-spread floored at the symbol's own decile spread, and a participation-based impact term:

$$c_{\mathrm{side}} = f_{\mathrm{taker}} + \tfrac{1}{2}\,\mathrm{spread} \cdot \max\!\big(1,\ \sigma_t/\bar{\sigma}\big) + k_{\mathrm{impact}}\,(Q/\mathrm{ADV}). \tag{4}$$

This model is a secondary diagnostic. **The headline nets a flat** 0.30% **round trip**, the pessimistic end of the pre-registered 0.1–0.3% band, pinned on the conformance surface and hard-checked before any cell is scored. Reporting the headline at the worst end of the band, and the model-based figure alongside it, is deliberate: a result that needs the optimistic end of its own cost assumption is not a result.

Positions are taken only when the forecast is large enough to be worth its cost. The execution filter is $\theta = \kappa c$: trade only when $|\hat{\mu}| > \theta$, with κ selected on the validation block from the grid $\{1.0, 1.5, 2.0, 3.0\}$ and then held fixed.

One measured property of this filter is worth stating, because it changes how a calibration result should be read. At the pinned horizon the threshold never bound on the fixture's degenerate checkpoints: for the three noise-arm cells re-scored there, the set of periods traded equalled the set scored — 12,000 of 12,000 — at every κ and at both volatility scalings tested, which makes the realized information ratio on those cells a pure function of $\mathrm{sign}(\hat{\mu})$ rather than of forecast magnitude. At shorter horizons the same checkpoints fail at the opposite endpoint, trading nothing at all: one of the three traded zero of 12,000 at $h \in \{1, 2, 3, 5\}$, and every noise checkpoint we hold is pinned to one endpoint or the other at $h \leq 3$. Both are the same pathology — a filter that never discriminates — and the guard below treats them alike. On a non-degenerate checkpoint of the *same* fixture it bound at every κ: the planted cell 4 traded 96.2%, 94.4%, 92.5% and 88.7% of scored decisions at $\kappa = 1.0, 1.5, 2.0, 3.0$. The two readings together identify the cause, and it is not the fixture. Binding is decided by the *forecast* magnitude, never by the realized return: on the non-binding cells $\hat{\mu}$ had collapsed to a near-constant per-step offset — a decode artifact, with |mean| between 23 and 283 times its own standard deviation across the three cells — so at $\kappa = 1.0$ it sat 4.4 to 31.6 thresholds away from zero while its own standard deviation was only 0.11 to 0.48 of one threshold, and cleared the filter everywhere. Raising the volatility scaling moved those cells *further* from binding rather than nearer, which is the signature of an offset rather than of a threshold set too low.

Whether the filter binds on real data is therefore an open question about the trained models, not about the design, and we do not assume the answer. Nothing in this repository measures it: no eval artifact from a real-lake run retains a decision-activity value, so the question is genuinely unanswered rather than answered elsewhere. What the pre-registration commits to instead is a refusal. The degeneracy guard reads each cell's decision-activity at κ^* and its fraction of negative forecasts, and returns HALT_ADJUDICATE — not a null — for any cell whose book is constant-direction. We record one asymmetry in that guard rather than discovering it later: the sign-balance leg carries a band, halting outside $[0.05, 0.95]$, while the filter-binding leg tests the exact endpoints only, so an activity of 0.999 passes where a fraction-negative of 0.999 halts. A near-degenerate book on the activity leg would not be caught, and the project has already measured a 0.99883 reading on the sibling leg, so the case is not hypothetical.

Forecasts are the conditional mean $\hat{\mu}$, not the mode. The estimator was changed from greedy-argmax decoding to an expectation decode after argmax was found to compound a sign bias over a rollout — fixture mean −0.051 against a standard deviation of 0.036, with frac-negative 0.929, versus −0.004 and 0.534 under expectation, while the correlation between $\hat{\mu}$ and the planted state *rose* from 0.858 to 0.898. The change is a correctness fix, made before any real cell was scored, and it re-anchored the reproducibility hash because the predictor sits inside the anchored evaluation instrument.

Information coefficient, rank information coefficient, mean absolute error and R^2 are computed and reported as secondary diagnostics. They do not enter the decision rule.

4.5 Evaluation geometry

Models are trained once and evaluated forward; nothing is refit per fold. Figure 6 shows the geometry. The four-year window is split at its 0.7 point, on 2023-10-20, into a training region and a forward region; the forward region is cut into six equal blocks. Block 0 is spent selecting κ and is excluded from every headline number, so the headline is scored on blocks 1–5, spanning 2024-01-01 to 2025-01-01 — a region the selection never saw.

Between any training window and a label that could contaminate it sit two margins: a 60-bar label look-forward, which is the maximum horizon the autoregressive rollout reaches, and a 60-bar serial-correlation margin. Their sum is a 120-bar leading embargo.

The premise behind the second margin — that 60 bars is enough for serial correlation to have decayed — is measured rather than assumed, on all 200 instruments, with a control arm that recovers a known AR(1) answer first so that a null reading is a measurement and not a silent failure. For signed returns, the channel through which a label leak would actually travel, the autocorrelation at lag 60 is 0.0067 on average and 0.0191 at the worst symbol, against a margin of 60: roughly a 24× margin. For absolute returns it is 0.2323 on average and 0.3604 at the worst symbol, and the premise does *not* hold. That is volatility clustering rather than a label-leakage channel, and Section 7 states what is and is not established by measuring only the signed channel.

Confidence intervals on ΔIR come from a paired moving-block bootstrap over the portfolio series, $B = 10{,}000$ resamples at a fixed seed, with $\lceil \sqrt{T} \rceil$ blocks and percentile intervals. The pairing is not cosmetic. Cells 4 and 5 hold the same market exposure, so the common factor cancels in the paired series; the average raw cross-sectional correlation is 0.6387, which leaves an effective breadth of about 1.5 symbols out of 40. The cross-section here buys robustness, not statistical power, and we say so rather than counting 40 independent bets.

4.6 The decision rule

The rule was written, timestamped and committed before any cell was trained. Figure 8 shows it as the runner evaluates it: five clauses, conjunctive, so the primary outcome is SURVIVES only if none fails and NULL otherwise.

1. **Paired interval.** The one-sided 95% lower bound of $\Delta\mathrm{IR}(4-5)$ exceeds zero.
2. **Detectability.** $\Delta\mathrm{IR}(4-5)$ reaches the paired minimum detectable effect, so an effect too small for this design to resolve cannot be reported as one.
3. **Placebo independence.** The one-sided lower bound of $\mathrm{IR}(4) - \mathrm{IR}(2)$ exceeds zero — a comparison that does not involve the placebo at all, so a placebo pathology cannot by itself produce a positive verdict.
4. **Economic materiality.** $\Delta\mathrm{IR}(4-5) \geq 0.5$ annualized information ratio, fixed before data. The rule string names the tested quantity as (information) + (capacity handicap), so the manifest cannot be read as claiming the information effect alone cleared it.
5. **Multiplicity.** The deflated Sharpe ratio is at least 0.95 over the enumerated trial set.

Two guards sit outside the clauses, and both are HALT-only: they can refuse to report an outcome their own inputs cannot support, and they can never turn SURVIVES into NULL or the reverse. The degeneracy guard fails closed if any cell–seed pair yielded non-finite forecast diagnostics, or if a required diagnostic is missing — a response to a defect in which the guard reported itself armed while examining nothing. The power guard compares each cell's across-seed range to the effect being claimed and halts if the wobble of the inputs reaches the size of the claim.

The outcome taxonomy has four words and all four are publishable: SURVIVES, NULL, INCONCLUSIVE, and HALT_ADJUDICATE. Committing to INCONCLUSIVE in advance is what makes the design honest under low power — the response to an underpowered result was written out and signed before the seed count was fixed, so "we did not have the power" is a pre-registered outcome rather than a post-hoc excuse.

Two further requirements are enforced at the manifest level rather than trusted to prose. A verdict manifest missing any required field cannot be quoted, and the loader refuses it; the required set includes a `dual_specification` field carrying the verdict computed under both the current and the pre-amendment specifications from the same artifacts, with any disagreement between them a first-class finding rather than a footnote. And every manifest carries a required disclosure, refused if absent: *we cannot exclude that our BSQ baseline is weaker than a reference BSQ implementation, which would inflate the FSQ-versus-BSQ comparison*.

4.7 Multiple testing and power

The deflated Sharpe ratio corrects for the number of configurations the design could have reported, and that number is enumerated rather than estimated: five cells × three horizons × four values of κ = 60 trials. The factorization is computed from named axes in code and asserted by a gate, together with the negative form proving the count does not silently acquire a seeds axis.

Seeds do not enter that count, and the reason is worth stating precisely because it looks like a convenience. The five seeds are *replicates of one configuration*, not five configurations we could have chosen to report. A multiple-testing adjustment cannot legitimately become stricter because the same specification was repeated more times; if it did, replication — which reduces variance — would be penalized as though it were search. The full 300-value cross-product of cells, seeds, horizons and κ is retained as an audit trail so that a reader can verify no seed-level selection occurred, but the correction is applied over 60.

The null dispersion the deflation uses is taken from the placebo cell alone. The earlier specification pooled all arms, which made this clause anti-correlated with its own hypothesis: a larger treatment effect inflated the pooled dispersion and made the clause harder to pass, to the point of requiring a cell-4 information ratio of 3–10 against an economic floor of 0.5. A proposed fallback to cell 2 was rejected because §5 can claim $\mathrm{IR}(2) - \mathrm{IR}(1)$, which makes cell 2 a treatment arm rather than a null.

The amendment was made before any data and its direction was recorded in advance as indeterminate: it loosens clause 5 by 0.859× while clause 2 tightens by 1.24–1.60×.

The minimum detectable effect carries two terms — scoring noise and training variance — and is computed with a self-written Student-t quantile and Welch–Satterthwaite degrees of freedom rather than a normal approximation, because at three to five seeds the degrees of freedom are small enough that the difference matters. The reference figure at the pinned horizon is 3.518 annualized information ratio.

We state the consequence plainly, in advance. That reference figure stands against an economic floor of 0.5 and a prior, measured on this data at an earlier milestone, of $|\text{RankIC}| \approx 0.027$ for trade-flow imbalance. The paired minimum detectable effect governs and will sit below the reference figure to the extent that cells 4 and 5 share exposure, by an amount not knowable before the data. **On these numbers, NULL or INCONCLUSIVE is the most likely honest outcome**, and nothing in the design manufactures power it does not have.

4.8 Pre-registration as the integrity mechanism

Everything above — the cells, the parity band, the placebo construction, the five clauses and their thresholds, the horizon, the cost, the κ grid, the seed count, the trial count and the null basis — is pinned on a conformance surface that the runner checks before it trains anything, and each pin carries a mutation test proving the gate rejects its predecessor value. Changing a pin is not a code edit that a reviewer must catch; it fails the run.

The pre-registration is a dated document with an append-only amendment log. Every amendment records what changed, when, why, and — where the direction was knowable — whether it made the bar easier or harder, recorded before the data that would reveal which. Amendments made after seeing calibration results are marked as such, including the one described in Section 3.6. The log also contains withdrawn claims: statements this project made and later retracted, retained rather than deleted. We regard that as the load-bearing part. A pre-registration that only ever records successful predictions is not evidence of discipline; one that records its own corrections is.

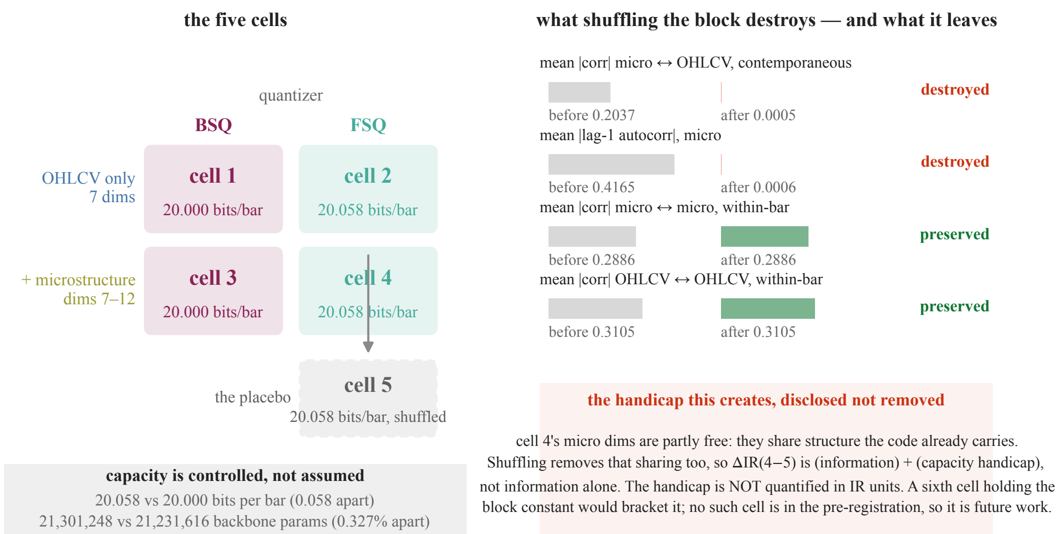

Figure 5: The five cells, at matched capacity, and the exact reach of the placebo. *Left:* a 2×2 over quantizer and input arm, plus a fifth cell that is cell 4 with the microstructure block shuffled. Capacity is controlled rather than assumed: the two quantizer arms differ by 0.058 bits per bar and by 0.327% of backbone parameters, so a difference between them cannot be attributed to one arm simply having more room. *Right:* what the shuffle actually does, measured. It destroys the two things it was built to destroy — microstructure's contemporaneous link to price ($0.2037 \to 0.0005$) and its own lag-1 autocorrelation ($0.4165 \to 0.0006$) — and leaves the within-block and within-OHLCV correlations exactly as they were. The consequence is stated rather than hidden: because cell 4's microstructure dimensions are partly amortized against structure the code already carries, shuffling removes that sharing too, so $\Delta\text{IR}(4{-}5)$ is (information) + (capacity handicap). The handicap is not quantified in information-ratio units. A sixth cell holding the block constant would bracket it, but no such cell appears anywhere in the pre-registration, so it is future work rather than a committed contingency, and Section 7 carries the handicap as an open disclosure.

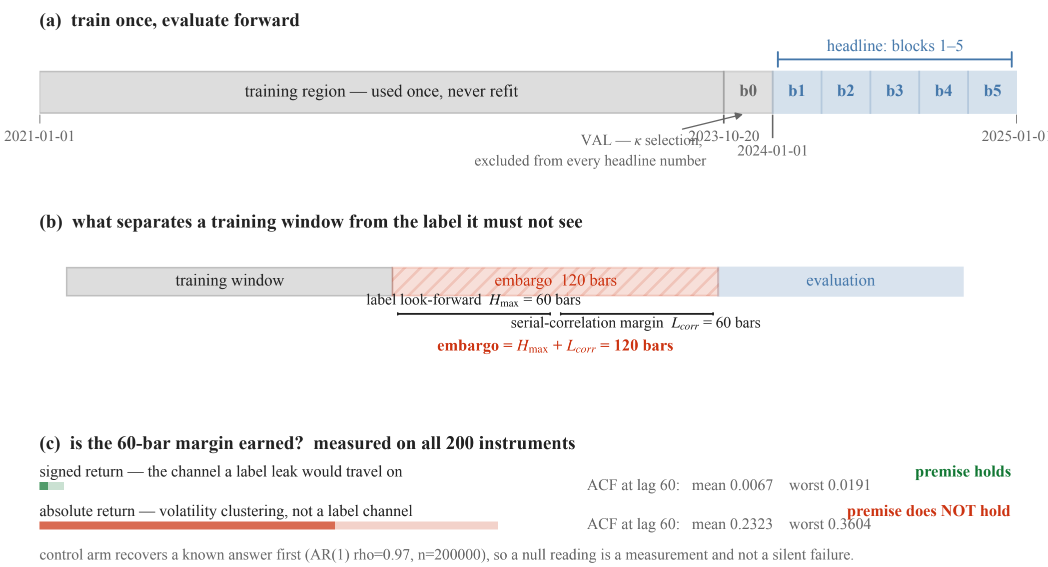

Figure 6: Train once, evaluate forward, and pay for the separation. **(a)** The training region ends at one boundary and is never refit. The forward region is cut into six blocks; block 0 is spent selecting the execution filter and is excluded from every headline number, so the headline is scored on five blocks the selection never saw. **(b)** Between a training window and any label that could contaminate it sit a 60-bar label look-forward and a 60-bar serial-correlation margin, giving a 120-bar embargo. **(c)** The premise behind that margin is measured on all 200 instruments rather than assumed, and a control arm recovers a known AR(1) answer first, so a null reading is a measurement and not a silent failure. It holds for signed returns — the channel a label leak would travel on — at ACF 0.0067 mean and 0.0191 worst-symbol at lag 60. It does *not* hold for absolute returns (0.2323 mean, 0.3604 worst), which is volatility clustering rather than a label channel; Section 7 carries that caveat.

5 Data

5.1 One venue, and the universe chosen from inside it

Every instrument is a USDT-margined perpetual future on Binance, sampled at one-minute frequency over the four years 2021-01-01 to 2025-01-01. Two public sources are combined per bar: the klines dump supplies open, high, low, close, volume and quote volume; the aggregated-trades dump supplies the signed trade statistics that the microstructure block is built from. No paid feed and no order book is involved, which is a scope limit (Section 7.7) and also the reason the corpus is reproducible by anyone.

The symbol set is bootstrapped *from the data* rather than from the exchange's live instrument list, and that choice is the one that decides whether the universe is survivorship-biased. A live instrument listing contains only instruments that still exist; selecting from it silently deletes every instrument that failed, which is the single most common way a backtest flatters itself. We instead scan the S3 index of the aggregated-trades dumps once, which yields per symbol, without downloading a single bar, the set of monthly dumps that exist. From that: the listing month is the first dump, and an instrument is treated as delisted when its last dump trails the newest global month. Ranking by total aggregated-trades bytes — a liquidity proxy available from the same index — and taking the top 200 gives the universe: **200 symbols, of which 41 were already delisted at the time of the scan**, each contributing only its [listing, delisting) bars.

One detail in that rule is worth stating because it changes three symbols. A dump index is published with lag, so an instrument that is merely a month behind the freshest global dump is not delisted — it is late. The delisting test therefore carries a one-month publish-lag tolerance, which corrected three symbols relative to a zero-tolerance comparison. Without it the universe would have carried three phantom delistings.

The realized ingest is **304,625,181** bars across all 200 symbols. That is below the 500M–1B band the blueprint nominated, and we flag it rather than quietly re-describe the band: 200 instruments over the fullest available window, many of them partial-coverage through late listing or delisting, lands near 300M. We regard it as adequate rather than merely acceptable, for a reason that is about the model and not about the data — a 21M-parameter backbone saturates well below this, so the lake is bought for regime and cross-sectional breadth rather than as training fuel. Reaching the nominal band would need roughly 320 instruments at proportional bandwidth, and would not train a better model.

5.2 What a bar is, and what three of its dimensions are not

Each bar is reduced to a 16-dimension feature vector. Seven dimensions are price shape and liquidity: log close-to-close return, log range, body fraction, upper and lower wick fractions, log volume and log quote volume. Six are the microstructure block, computed from aggregated trades: trade-flow imbalance $(v_{\text{buy}} - v_{\text{sell}})/(v_{\text{buy}} + v_{\text{sell}})$, signed count imbalance, log trade count, log mean trade size, trade-size dispersion, and large-trade share. These six — indices 7 through 12 — are the block the ablation varies, and the first of them is the quantity the paper is about.

The remaining three dimensions — funding rate, log open interest and its change, indices 13 to 15 — are specified, wired through every transform, and **structurally absent**. The v1 ingest covers klines and aggregated trades only, so on all 304,625,181 bars those three columns have standard deviation exactly 0.0 and their masks are set on 100% of rows. We state this here rather than in a footnote because the natural reading of "a 16-dimensional microstructure vector for perpetual futures" is that funding and open interest are in it, and they are not. The vector is 16 wide and 13 live. Section 7.7 carries what that costs.

Every feature carries a companion mask bit, and the mask travels with the value through every downstream transform. That pairing is not bookkeeping: a defect found in review had the placebo shuffle permuting values while leaving masks in place, which would have stranded fill values on valid bars and real values on masked ones. It was fixed before any training run.

5.3 Normalization that cannot see forward

Feature scales in this market move by orders of magnitude across four years, so the unbounded dimensions are normalized rather than used raw. The normalization is where lookahead usually enters a financial pipeline, through a statistic fitted on the whole sample and then applied to its own training region, so the rule here is absolute: every statistic used for bar t is computed from data available at the close of bar t.

Nine dimensions — the unbounded, scale-varying ones — receive a causal exponentially weighted z-score, at a half-life of 1,440 bars (one day) for the fast group and 5,760 bars (four days) for the slow volume and open-interest group. Seven dimensions are already bounded or relative, and are clipped only. Everything is clipped last to $[-5, +5]$ as an outlier guard. The volatility scale σ_t that the prediction targets are expressed in is built causally from raw returns within a segment, on the same rule.

Each statistic needs a warm-up before it is trustworthy, and warmed-up bars are dropped rather than back-filled. The cost is small and it is measured, not estimated: on the three thinnest instruments in the universe the warm-up drop is 809 of 279,930 bars, 811 and 809 respectively — about 0.3% — because most instruments are long single segments and the warm-up is a one-time tax at the start of each.

Causal safety is not asserted, it is a required continuous-integration gate. The test suite plants lookahead leaks of several kinds — a global one, a localized divisor leak, and a survivorship leak in the target-validity flag — and asserts that the exhaustive sweep catches each. One of those tests exists specifically to show that a *sampling* check misses the localized leak that the exhaustive check catches, so the gate cannot be quietly weakened to a sampler. During ingest the same sweep ran in-line on real cross-sections and a red result raised a fatal error before anything was recorded.

5.4 What the quality gates cost

Of the 304,625,181 bars, 303,283,972 (99.56%) **are usable by the OHLCV-only arm** after warm-up and quality masking, and 300,311,536 (98.58%) **are usable by the microstructure arms**. The ratio between those — microstructure availability given price availability — is 99.0% universe-wide, and no instrument falls below 0.7. Across the universe, 0.175% of bars are stale and 0.376% have zero volume.

That second number is the one that matters for the placebo. If microstructure were missing on a large fraction of bars, cell 4 and cell 5 would differ in coverage as well as in content, and the comparison the paper rests on would be confounded by absence rather than by information. At 99.0% availability with no starved instrument, it is not.

Forty-three instruments are thin, by a joint rule — fewer than half the median usable bars, or more than 10% stale. Inspecting them rather than dropping them: almost all are recent 2024 listings with 280k–500k bars and microstructure availability near 1.00, so they are thin by *age*, not by quality. The single genuinely illiquid instrument is a delisted one at 2.5% stale and 2.6% zero-volume. All 43 remain in the lake; thinness is handled as a sampling weight in the training draw, not as a deletion.

5.5 Coverage, and the survivorship we did not remove

Figure 7 is the whole universe as a symbol-by-month raster: 7,024 of a possible 9,600 symbol-months are present. The left staircase is instruments listing at different dates. The right edge is ragged because instruments that stop trading are *kept* in the lake, and their absence afterwards is real absence rather than a filter. A survivorship-cleaned universe renders as a solid rectangle, which is precisely the artefact the panel exists to demonstrate we did not create. Two instruments have interior gaps rather than truncated tails and appear as internal white streaks.

Two delisting counts appear in this project and they count different things, so we state both. The ingest record's **41** is the number of instruments known to be delisted as of the index scan, at any date. The **17** in Figure 7 is the number whose data ends *inside* the four-year window, which is all a raster over that window can show; the remaining 24 delist after 2024-12 and are complete here. The figure and this section therefore say "stop trading inside the window" and never "delisted".

Survivorship is owned, not solved. Retaining failed instruments removes one bias; it does not remove listing bias, since an instrument must have been listed on this venue at all to enter the universe.

Section 7.5 carries that.

5.6 Storage, provenance, and what makes a past prediction reconstructible

The reduced lake is Hive-partitioned Parquet under `(symbol, frequency, month)`, queried through DuckDB. It was first written per-day, at 211,595 files, which made per-symbol globbing pathological; compaction to one file per symbol-month gives 7,024 files and 15 GB. The compaction was proven content-preserving rather than assumed: all 200 per-symbol dataset hashes were re-derived from the compacted lake and the universe Merkle root recomputed, with zero mismatches and an unchanged root. Identical features, masks and timestamps imply an identical causal result, so the hash comparison subsumes a re-run of the causal sweep.

Every dataset is content-hashed and the universe carries a single Merkle root over the lake, `sha256:5dfd667d05b97bda`..., recorded in the manifest alongside per-symbol listing and delisting dates and a corpus-purpose note.

Raw dumps are not retained, and that is a deliberate position rather than a shortcut. The Merkle root and the per-symbol hashes are taken over the *reduced* lake, which is the artifact we keep; the Binance SHA-256 of every raw shard is recorded for provenance only. Re-pulling raw and re-running a deterministic reducer reproduces the recorded lake hash — the reducer was proven byte-identical at the preceding milestone — so reproducibility survives eviction of 503 GB of raw input. Section 8 states exactly which parts of the pipeline are bit-exact and which are not.

5.7 What the experiment reads

The lake is broader than the experiment. Of the 200 instruments, the headline metric is scored on 40; the four-year window is split at its 0.7 point, on 2023-10-20, into a training region and a forward region, and the forward region is cut into six blocks of which block 0 is spent selecting the execution filter and excluded. The headline is therefore computed on blocks 1 through 5 — roughly 2024-01-01 to 2025-01-01, one year, 35,063 stride-15 periods per instrument — on a region no selection step has seen. The other 160 instruments and the preceding three years are lake breadth and training data, not evaluation breadth, and Section 7.5 states what that leaves unestablished.

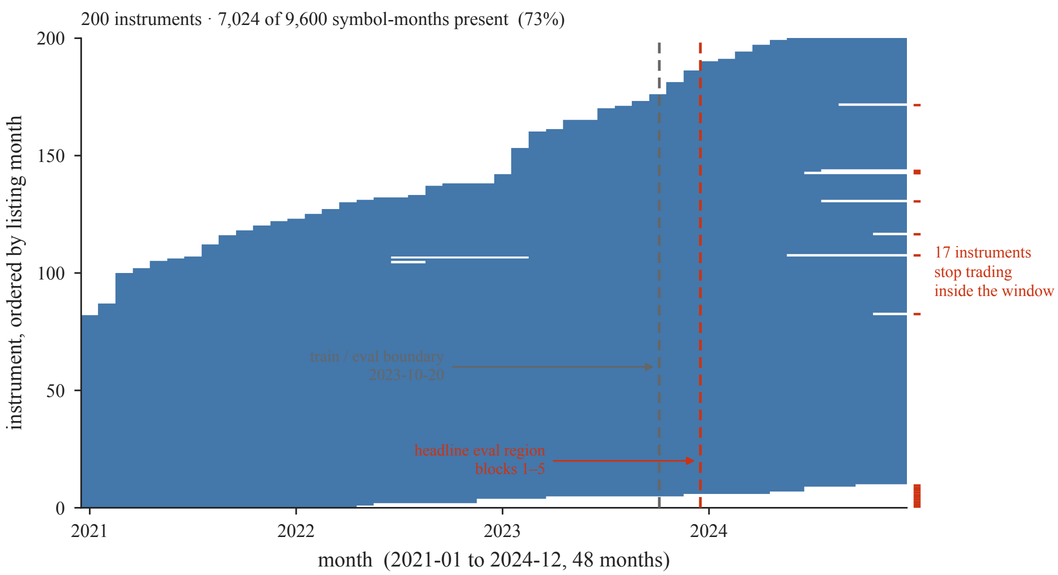

Figure 7: Survivorship is owned, not filtered away. Month-by-month presence for all 200 instruments over the four-year window, ordered by listing month: 7,024 of a possible 9,600 symbol-months are present. The left staircase is instruments listing at different times; the right edge is ragged because instruments that stop trading are *kept*, and 17 of them stop inside this window. A survivorship-filtered universe would render as a solid rectangle, which is the artefact this panel exists to show we did not create. Two instruments have interior gaps rather than truncated tails and appear as internal white streaks. The two dashed rules are the committed fold plan: training ends at the first, and the headline metric is scored only to the right of the second. Note that 17 counts instruments whose data ends inside the window; the ingest record's 41 counts every instrument known to be delisted at any date, including 24 that delist after this window closes.

the pre-registered decision surface conjunctive — SURVIVES only if none fails

		outcome
1 paired ci	one-sided 95% CI lower bound of ΔIR(4−5) $>$ 0	—
2 mde paired	ΔIR(4−5) $\geq$ paired MDE (tabled 3.518 at h=15)	—
3 placebo validity	CI lower bound of IR(4)−IR(2) $>$ 0 (placebo-independent)	—
4 economic floor	ΔIR(4−5) $\geq$ 0.5 annualized IR	—
5 dsr	deflated Sharpe ratio $\geq$ 0.95 over 60 enumerated trials	—

PLACEHOLDER — awaiting the run

HALT-only guards — never flip SURVIVES ↔ NULL
degeneracy — power —

SURVIVES / NULL / INCONCLUSIVE / HALT_ADJUDICATE
emitted —

Figure 8: The decision surface, fixed before any result existed. Five clauses, evaluated conjunctively: the primary outcome is SURVIVES only if none fails, and NULL otherwise. Two guards sit outside the clauses and are HALT-only — they can refuse to report an outcome their own inputs cannot support, and can never flip SURVIVES to NULL or back. The rule strings shown are the ones persisted into the verdict manifest, not a restatement of them. **[PH: no values]** — every outcome cell is empty because the run has not been executed; each is filled from a single named field of that manifest.

6 Results

[§6 — structured stubs only. No number below exists until the pre-registered run completes; the three figures are built against the artifact schema and are watermarked until then. Both interpretation paragraphs are pre-written and dated.]

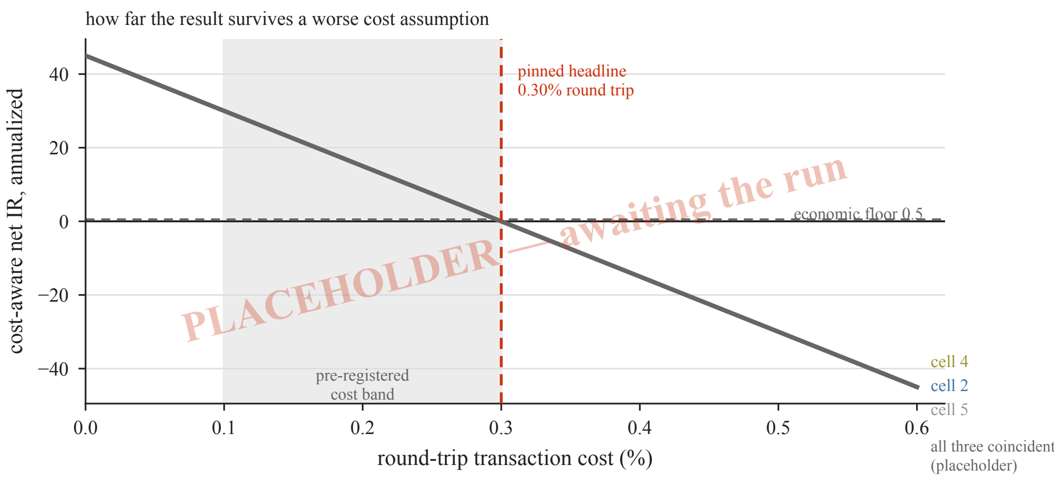

Figure 9: How far the result survives a worse cost assumption. Cost-aware net information ratio against the round-trip transaction cost, for the microstructure arm, the OHLCV-only arm and the placebo. The headline is pinned at 0.30% — the pessimistic end of the pre-registered 0.1–0.3% band — so the region to the right of that rule is stress beyond what was committed to, and the crossing of the economic floor is the quantity a reader should look for first. **[PH: one coincident curve, magnitudes ∼ ±45]** — all three cells are drawn on a single line at a scale roughly two orders of magnitude above anything this design could produce, so no ordering between cells and no crossing point can be read off. The axes, the pinned rate and the floor are real; nothing else on the panel is.

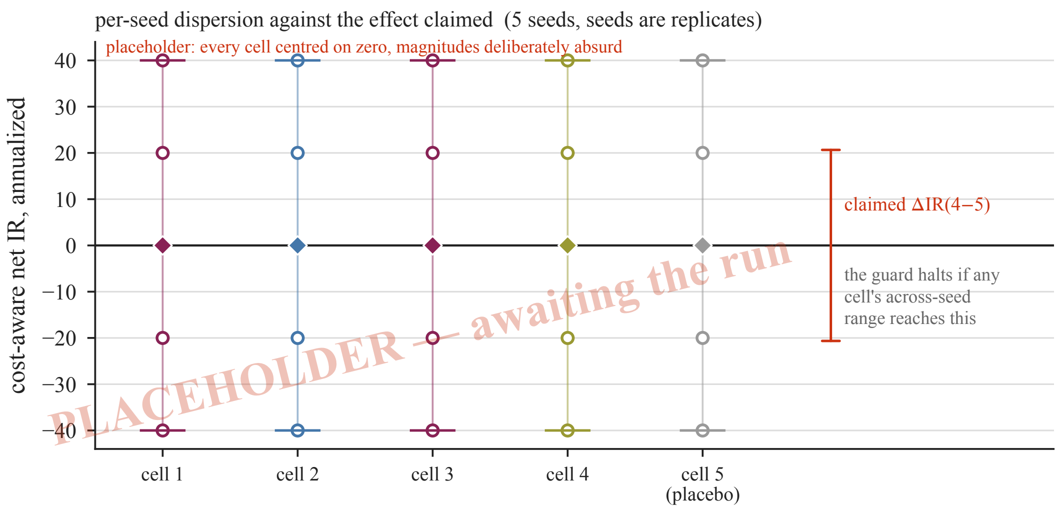

Figure 10: Seed dispersion against the effect being claimed. Each cell's five seeds are replicates of one configuration, not five configurations; open circles are seeds, the diamond is the seed mean, and the bars mark the across-seed range. The comparison that matters is the one drawn on the right: if any cell's across-seed range reaches the size of the effect being claimed, the power guard halts and the verdict is reported as adjudication-required rather than as an outcome. **[PH: every cell centred on zero, magnitudes** $\sim \pm 40$**]** — no cell is favoured and the claimed effect is drawn symmetric about zero, so the panel shows the comparison's geometry and no result.

7 Limitations

The limitations below are stated at the same confidence as the contributions, and in the order a sceptical reader should apply them. Three of them — the synthetic basis of the mechanism, the absent external baseline, and the placebo's capacity confound — bound what the paper is entitled to claim even if every pre-registered clause passes.

7.1 The mechanism is measured on a synthetic fixture, not on markets

Everything in Section 3 is measured on a constructed fixture. That is a deliberate trade and it cuts both ways. The fixture's non-signal structure is entropy-matched to the real lake — a per-dimension deficit of 0.4631 nats, reproduced at 0.462 against a declared tolerance of 0.05 — and its signal dimension is planted with an exactly computable information content, which is what allows us to say that 94.4% of a known quantity was recovered rather than that a loss curve moved. It is also precisely what stops the result being a measurement of markets. We do not know the variance and covariance geometry of real trade-flow imbalance against real price well enough to assert that the fixture reproduces it; we know only that one entropy statistic matches.

The consequence is a specific one, and we hold to it under every experimental outcome: capacity eviction is a mechanism we demonstrate, not a market phenomenon we measure, and it is not this paper's headline claim. A reader who wants the mechanism established on real microstructure will not find it here. What the pre-registered ablation of Section 4 tests is the downstream question — whether real trade-flow imbalance survives the re-specified pipeline — and a positive result there would be consistent with the mechanism without confirming it, because many other differences between cells 2 and 4 could produce the same sign.

Two narrower limits inside the same fixture. We demonstrate the effect for one reconstruction objective, one quantizer family and one class of encoder. We show that *this* objective allocates by variance and covariance; we do not show that no reconstruction objective can do otherwise, and Section 3.6 is a counterexample to the strong form — it is a reconstruction objective that carries the evicted channel, because it was given an explicit term to do so. Separately, the 94.4% recovery figure is calibrated against a single matched reference run and inherits whatever run-to-run variation that reference carries; we report it as one measurement, not as an expectation over runs.

7.2 The BSQ baseline is ours, and nothing external anchors it

Cell 1 — binary spherical quantization on OHLCV-only inputs — is our own implementation at matched bits per token. It is not an independently validated reference. The evaluation harness was originally specified to load published reference weights and reproduce their reported rank information coefficient as an external anchor for this cell. That gate was found unexecutable and dropped as binding before any cell was scored, for three separate reasons, each sufficient on its own: the reference paper publishes two rank information coefficients for the same model that differ by 2.4× (0.0431 against 0.0665) without the pre-registration saying which applies; both are measured on Shanghai Stock Exchange 15-minute equity bars, a market and a frequency this study firewalls out of v1, so the gate's two requirements — reproduce the published number, and do it on our own pinned slice — cannot both hold; and executing the weights would require importing the reference model code, which our independence invariant forbids. Resolvability was never the obstacle: the band is 4.51 to 11.05 standard errors wide at the measured effective sample size of 1,401,637.

Two compensating controls were armed in its place, and it is important to be exact about what they do. A saturation check asks whether cell 1 was trained enough to be a fair baseline. A codebook utilization gate, pinned at 0.95, asks whether our BSQ codebook collapsed. Neither answers the question the external gate was for. We therefore carry the following sentence as a required field of every verdict manifest, refused if absent, and reproduce it verbatim here:

> We therefore cannot exclude that our BSQ baseline is weaker than a reference BSQ implementation, which would inflate the FSQ-vs-BSQ comparison reported in Section 6.

This bounds the quantizer leg specifically. It does not bound the input-arm leg: the comparison between cells 2 and 4, and between cell 4 and the placebo, is internal to our own implementation on both sides, and a weak baseline cannot inflate a difference in which the same baseline appears twice.

7.3 The placebo removes information and capacity together

The shuffled-microstructure cell is the design's central control, and its reach is narrower than its name suggests. Measured on the fixture, the shuffle destroys exactly what it was built to destroy: microstructure's contemporaneous link to price falls from 0.2037 to 0.0005, and its own lag-1 autocorrelation from 0.4165 to 0.0006, while the within-block and within-OHLCV correlations are left as they were.

But microstructure dimensions in cell 4 are partly amortized against structure the code already carries, and shuffling removes that sharing along with the information. $\Delta\mathrm{IR}(4-5)$ is therefore (information) + (capacity handicap), and **we have not quantified the handicap in information-ratio units**. A sixth cell holding the microstructure block constant rather than shuffled would bracket it, and no such cell appears anywhere in the pre-registration — we have verified this by exhaustive search of that document — so it is future work, not a committed contingency we chose not to exercise. The practical consequence: a positive $\Delta\mathrm{IR}(4-5)$ is evidence that microstructure carries information beyond input capacity, and is not a calibrated estimate of how much.

7.4 The effect we can detect is larger than the effect we expect

At the headline horizon the paired minimum detectable effect is 3.518 annualized information ratio, against an economic materiality floor of 0.5. The only prior we hold for the quantity under test is our own univariate liveness screen of trade-flow imbalance on a single symbol: $|\mathrm{RankIC}| = 0.0270$ at $h = 5$ and 0.0197 at the headline $h = 15$, both with confidence intervals excluding zero. That screen's own source describes it as a liveness screen and not the headline test, and we use it here only to set expectations about magnitude. An effect of that size does not become an information ratio of 3.5. We therefore state plainly, before seeing any result, that a null or inconclusive outcome is the most likely honest one, and that this was pre-committed as publishable: the outcome taxonomy has three members, and the decision rule names what we report under each.

A second, less visible power limitation. The minimum detectable effect above is built from *scoring* noise — the dispersion of the portfolio return series — and contains no term for training variance. That omission is not conservative, and we found its magnitude by accident: two runs at the same seed, the same recipe, the same horizon and the same estimator produced fraction-negative 0.5662 and 0.99883 for the same cell. Same-seed divergence is basin-hopping, and it is a lower bound on across-seed variance, which is the quantity the design actually averages over. Five seeds are pinned up front, which helps; forcing deterministic kernels does not help, because it removes the same-seed component and leaves the across-seed component untouched.

What we do about it is a refusal rather than a correction. A power guard, computed after the clauses and unable to alter them, compares each cell's across-seed range against the effect being claimed; if the range reaches the claimed effect, the verdict is reported as HALT_ADJUDICATE rather than as an outcome. The guard is HALT-only by construction — it can manufacture neither a positive nor a negative claim — which is what makes it legitimate to have added it before the run. It does not increase power. It prevents us from reporting a number smaller than its own inputs' wobble.

7.5 One exchange, one venue, one year of scored data

The lake is 304,625,181 one-minute bars across 200 instruments, and the breadth of that number should not be read as breadth of evidence. Every instrument is a USDT-margined perpetual future on a single exchange. Perpetuals are not spot, and one exchange is not the market: fee schedules, tick sizes, liquidation mechanics and the composition of flow all differ across venues, and trade-flow imbalance is a venue-local quantity in a way that returns are not. Nothing here establishes that the result transfers to spot, to another exchange, to dated futures, or to any other asset class.

Within that universe the scored slice is narrower still. The headline metric is computed on 40 instruments, not 200, and on forward blocks 1 through 5 — roughly 2024-01-01 to 2025-01-01, one year — with block 0 spent selecting the execution filter and excluded. The remaining 160 instruments are lake breadth, not evaluation breadth. One year of scored data at a 15-bar horizon is 35,063 periods per instrument, and the effective breadth across a correlated 40-instrument cross-section is 1.5, not 40 — which is where most of the minimum detectable effect in Section 7.4 comes from. That single year is also one market regime. We do not test regime transfer and cannot claim it.

Survivorship is owned rather than filtered: instruments that stop trading are retained, and 17 of them stop inside the window, giving the ragged right edge of Figure 7. This removes one bias; it does not remove listing bias, since an instrument must have been listed on this venue to enter the universe at all.

7.6 The embargo premise is established for one channel only

The 120-bar leading embargo is a 60-bar label look-forward plus a 60-bar serial-correlation margin, and the premise behind the second margin was measured on all 200 instruments rather than assumed, with a control arm that recovers a known AR(1) answer first so that a null reading is a measurement and not a silent failure. For signed returns it holds comfortably: autocorrelation at lag 60 is 0.0067 on average and 0.0191 at the worst instrument.

For absolute returns it does not hold: 0.2323 on average and 0.3604 at the worst instrument, at the same lag. We report this as a red finding rather than a footnote. Our position is that signed returns are the channel a label leak would actually travel on — the labels are signed forward returns, and a leak means a training window seeing the sign of a future return it should not — and that persistent absolute-return autocorrelation is volatility clustering, a well-established property of the series that is present in the training region and the evaluation region alike and is not a path from label to feature.

That is an argument, and it is the weakest link in the validation geometry. What we have established is that the signed channel is clean at the chosen margin. What we have *not* established is that no leakage path exists through volatility — for instance, a model that learns to condition on realized volatility could in principle be advantaged by a training window whose volatility state persists into the evaluation region. Extending the margin until the absolute-return premise also holds would require a margin several times longer and would cost evaluation data we do not have; we chose the shorter margin and we are stating the choice.

7.7 The microstructure leg is narrower than "microstructure" suggests

Trade-flow imbalance, not order-flow imbalance. The imbalance signal is computed from freely available aggregated trades, and is therefore signed *executed-volume* imbalance. True order-flow imbalance requires order book depth — the arrival and cancellation of resting liquidity — which we do not have. The distinction is not cosmetic: order-flow imbalance carries information about intent that never executes, and a substantial part of the microstructure literature's predictive content comes from exactly that. We use the name trade-flow imbalance everywhere, in code, configuration and prose, and a negative result here is evidence about trade-flow imbalance only. Depth-based order-flow imbalance is explicit v2 work.

Three of sixteen feature dimensions are structurally absent. The per-bar feature vector is specified as 16 dimensions, of which 13 are live. Funding rate and open interest occupy dimensions 13 through 15; they are wired, carried through the pipeline, and masked — and in the v1 lake they have standard deviation exactly 0.0 and are masked on 100% of all 304,625,181 bars, because ingest covers klines and aggregated trades only. They are specified and structurally absent. The microstructure block that the ablation actually varies is six live dimensions, 7 through 12, each non-degenerate across all 200 instruments. Funding and open interest are the two channels most plausibly carrying the information a perpetual-futures microstructure study would want, and the study is run without them. This is a limitation of the v1 data, not a design choice defended on merit.

7.8 Two matching caveats

λ is fitted to a synthetic proxy. The class weight that makes the per-bar bottleneck carry the microstructure block was selected on the calibration fixture, not on real data, and its acceptance threshold was restated after calibration results were seen — we state that plainly in Section 3.6 rather than let it be discovered. It is pinned pre-data at a single value and applied identically to every arm, so it cannot favour any cell relative to any other. The residual risk is that the proxy does not transfer: a λ tuned for a synthetic covariance geometry may under-serve the real one. The guard is the legibility gate, which is why it exists — it is evaluated on real data before any second-stage training begins, requires at least 0.90 sign accuracy on every one of the six live microstructure dimensions with no mean clause and no floor, and raises a hard stop rather than degrading quietly. A gate that fires costs us the run; it does not cost us the honesty of the result.

The arms are matched to 0.327%, not exactly. The backbone realizes 21,301,248 parameters at the FSQ vocabulary and 21,231,616 at the BSQ vocabulary — 69,632 fewer, or 0.327% — because embedding and output-head rows scale with vocabulary size. The two quantizer arms also differ by 0.058 bits per bar. Both gaps favour the FSQ arm, both are an order of magnitude smaller than any effect the design can detect, and both are reported rather than absorbed. We do not claim exact parameter parity; we claim parity to a stated tolerance, hard-checked before any cell is scored.

7.9 Reproducibility is bit-exact except in one place, and that place is training

The data pipeline, the frozen normalization statistics and prediction replay are bit-exact from one configuration file, a pinned seed and content-hashed inputs. GPU training is not, unless the deterministic-attention fallback is enabled, and the two are not the same claim: deterministic attention is necessary for bit-exact training and is not sufficient, because reduction order elsewhere in the backward pass is unconstrained. We record the mode for every run. Section 8 states exactly which artifacts carry which guarantee, and Appendix E separates what was measured from what was estimated across the whole paper.

8 Reproducibility and verification

8.1 What is bit-exact, and what is not

Reproducibility here is a deliverable rather than hygiene, and it is scoped rather than asserted whole. Three things are bit-exact from one configuration file, a pinned seed and content-hashed inputs: the data pipeline, the frozen normalization statistics, and prediction replay. Any recorded prediction can be reconstructed from the recorded inputs.

GPU training is not bit-exact by default, and the honest version of that sentence took us a correction to write. The production path now sets deterministic algorithms explicitly, at a measured throughput cost of 13.175%, and every run records the mode it ran in.

The correction is worth stating because it is the kind of error a reproducibility section normally hides. For a period, 49 of 49 run records carried an assertion of bit-exactness beside a configuration that had determinism switched off. The cause was a single line that computed the claim as *attention mode equals the deterministic kernel* — our own invariant's sufficiency premise encoded as code. Deterministic attention is necessary for bit-exact training and is not sufficient, because reduction order elsewhere in the backward pass is unconstrained. The scope of the false claim was established at the time and is narrow: only the GPU-training clause was affected; the pipeline, frozen-statistics and prediction-replay clause held, the anchor of Section 8.2 held, and no prior result was invalidated. We report it here rather than quietly shipping the fix because a reader assessing our other claims should know that this one was wrong for a while and how it was caught.

One consequence survives the fix. Same-seed runs were observed to diverge — fraction-negative 0.5662 against 0.99883 for the same cell at identical horizon, cap, estimator, seed and recipe — which is basin-hopping, and a lower bound on across-seed variance. Forcing determinism removes the same-seed component and leaves the across-seed component untouched, so it does not repair the statistical consequence. Section 7.4 carries that; Section 8.6 describes what we do about it.

8.2 The anchor

A reproducibility claim that is only checked at the end is a claim about one moment. We instead carry a standing anchor: a fixed evaluation of a frozen checkpoint through the full harness, whose results hash is re-proven before and after any change to the evaluation instrument. The current anchor is `sha256:3f86882a63dd06c78…`, over inputs that are themselves content-hashed — dataset, tokenizer and predictor each carry their own SHA-256 — and it is accompanied by an exhaustive causal sweep at 420 of 420 checks.

Two procedural rules attach to it, both of which exist because of a specific failure. The anchor is re-proven *twice*, from separate invocations. And its exit status is read from an unpiped command, because an exit code read through a pipe is the pipe's — a ‘`| grep | tail`‘ once masked a nonzero exit and let a false anchor claim into a commit message. The anchor has been legitimately re-based once, when a file inside the anchored instrument changed for a correctness fix; the re-basing is recorded rather than presented as continuity.

The environment is pinned by a lockfile: Python 3.11.7, NumPy 2.4.6, PyTorch 2.12.1. That pin is itself a response to a defect — environment drift with no lockfile — and load-bearing results are re-verified under it rather than under whatever interpreter happened to be first on the path.

8.3 Content addressing

Every produced dataset is content-hashed, and the universe carries a single Merkle root over the lake, recorded in the manifest with per-symbol listing and delisting dates (Section 5.6). The property this buys is specific: any past prediction can be reconstructed, because the exact bytes it was computed from are identified rather than described. The compaction of the lake was proven content-preserving by re-deriving all 200 per-symbol hashes and recomputing the root, rather than by inspecting the compactor.

8.4 Pre-registration, and an amendment log that records the direction

The decision surface — clauses, thresholds, horizon, cost, the κ grid, the seed count and the trial count — was fixed before any of its inputs existed, and the design was frozen before the run. Amendments are not forbidden; unlogged amendments are. Every change is recorded with its date, its reason and, crucially, a *direction-blind* test: before looking at whether the amendment helps, we record whether it tightens or loosens the criterion. The log carries 41 distinct revision tags.

The direction test is not decorative, and it has already overturned our own summary of our own amendments. The most consequential window changed the multiple-testing trial count and the dispersion basis for the deflation benchmark together, and it was written up as *every amendment in this window loosens the effective bar*. That was false. The deflation clause does loosen, by a factor of 0.859. But the clause that most directly tests the headline contrast *tightens*, twice over: its quantile multiplier rises from 2.4865 to 3.0728 at five seeds, a factor of 1.236, and the total standard error gains a training-variance term that can only increase it. The net across a conjunctive gate is indeterminate before the run and is not monotonically loosening. Had the original framing reached this paper it would have been a false statement about our own amendments, in the section a referee reads most carefully. We report the aggregate as one number rather than letting a favourable half stand for the whole.

Where a specification change means two defensible analyses exist, both are computed from the same artifacts and both are reported. `dual_specification` is a required field of the verdict manifest, and a manifest that lacks it cannot be quoted; disagreement between the two specifications is a first-class finding rather than a footnote.

8.5 The conformance surface

Pre-registration is only binding if something checks it. Every pinned quantity is a named constant that is hard-asserted before any cell is scored: the symbol set by hash, the κ grid, the five seeds, the flat headline cost of 0.0030, the evaluation sequence length, the realized backbone parameter count of 21,301,248, the $\hat{\mu}$ estimator, the training step counts, the bootstrap configuration, the causal-encoder flag, the pointwise-fine flag and the microstructure class weight. A run whose configuration differs from any of these does not produce a downgraded result; it raises and produces none.

8.6 Guards that refuse rather than decide

Two guards sit outside the decision rule, and their design constraint is that neither can change an outcome — each can only withhold one. The degeneracy guard refuses a verdict for any cell whose book is constant-direction. The power guard refuses one when a cell's across-seed range reaches the size of the effect being claimed. Both are computed after the clauses, neither mutates them, and neither can flip a survival to a null or the reverse.

That asymmetry is what made it legitimate to add them after the design was frozen: a guard that can only produce HALT_ADJUDICATE can manufacture neither a positive result nor a negative one. It is also why they are described here rather than in Section 4 — they are part of how the result is verified, not part of what is being tested. Section 4 states one asymmetry we chose not to remove and the reason.

8.7 How the defects were actually found

We keep a defect ledger, and we report it because the distribution in it is more informative than any assurance we could offer. It records 36 defects: 19 we classify as critical — meaning they would have produced a wrong published result — 15 high, and 2 medium. All but one are closed; the exception is the capacity-eviction finding itself, which is mitigated rather than closed because it is a property of the objective and not a bug.

The useful column is not severity but *how each was found*:

detection layer	defects	of which critical
Independent audit (first pass)	9	6
Builder or supervisor, outside a formal pass	8	3
Independent re-audit	6	2
Execution (found by running it)	5	2
Design review	4	2
Planted-signal canary	3	3
Independent audit (third pass)	1	1
Total	36	19

No single layer found even a third of the critical defects. The largest contributor, the first independent audit, accounts for 6 of 19. Three critical defects were found only by planting a signal of known information content and observing that it was not recovered — no review of that code would have found them, because the code was correct and the *interface* was not. Two were found only by executing the pipeline, which is the basis for the operational rule that a component whose first execution is on rented hardware is an untested component whatever its coverage says.

The reason to publish this table is that it is the strongest available argument against the verification story a paper normally tells. Code review, a green test suite and a careful author would have caught roughly a third of what mattered here. We do not claim the remaining defects are gone; we claim that seven independent detection layers were run, that each found something the others missed, and that the count of defects found by the most recent layer has not yet reached zero.

A recurring pattern is worth naming, because it changes what a reader should count as evidence. Several defects were checks that could not fail: a mutation test whose fixture made the two cases it separated numerically identical; a gate that shipped below a `return` and had never executed; a unit test whose written contract was the defect. In each, a test existed and was green. Our operating rule is therefore that a gate or negative test is suspect until it has been observed to fail, and that closing a finding requires restoring the pre-fix source in an isolated tree and watching the new test fail there.

8.8 What is released, and what a reader can check without it

The artifact is the code, the trained weights, the pre-registration with its full amendment log, the run manifests, and the scripts that regenerate every figure in this paper from those manifests at render time. No number in any figure is transcribed; each is loaded from the receipt that produced it, so a figure that disagrees with its artifact fails to build rather than rendering a stale value.

Two things a reader can check without rerunning anything: the anchor hash, which fixes the evaluation instrument, and Appendix E, which separates every claim in this paper into measured and estimated, with the artifact path for each. Section 7 states what remains unestablished.

References

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Aäron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Yue Zhao, Yuanjun Xiong, and Philipp Krähenbühl. Image and video tokenization with binary spherical quantization. *arXiv preprint arXiv:2406.07548*, 2024.

A Eviction measurement, per dimension

Table 2 gives the per-dimension point-decoder correlations summarized in Table 1. Marginal standard deviations are implied by each dimension's predict-the-mean baseline through $\mathbb{E}|X| = \sigma\sqrt{2/\pi}$, and confirm the fixture's construction: every dimension except the return is unit-variance to two decimal places, so the return is the only high-variance channel and the signal dimension is distinguished from the fillers by dependence structure alone.

Table 2: Point-decoder correlation for every fixture dimension, three seeds. Recovery of a dimension's value at bar t from bar t's own fine code. The eleven fillers are coupled by an AR(1) chain across dimensions at $\rho = 0.7993$; the signal dimension is independent of everything. Seed-to-seed variation within a class is at most 0.006, while the gap between the filler class and the signal dimension is more than 0.80.

dim	role	marginal sd	seed 0	seed 1	seed 2
0	return (high variance)	3.15	**0.9818**	**0.9815**	**0.9823**
1	filler	1.00	0.8931	0.8903	0.8920
2	filler	1.01	0.9128	0.9113	0.9109
3	filler	1.00	0.8523	0.8509	0.8565
4	filler	1.00	0.8261	0.8256	0.8319
5	filler	1.00	0.8793	0.8808	0.8816
6	filler	1.01	0.9120	0.9166	0.9101
7	filler	1.00	0.8793	0.8790	0.8738
8	filler	1.01	0.8258	0.8235	0.8242
9	signal (independent)	1.00	**0.0140**	**0.0011**	**0.0053**
10	filler	1.01	0.8513	0.8486	0.8514
11	filler	1.00	0.9179	0.9188	0.9171
12	filler	1.01	0.8969	0.8975	0.8940

Four dimensions clear a point-decoder correlation of 0.90 in each of the three seeds, which is the number the capacity arithmetic of Section 3.1 anticipates: a 10.26-bit fine budget cannot support thirteen dimensions at the fidelity that a 0.90 sign accuracy requires.

B Token-space plant: construction and calibration

The control of Section 3.4 requires a plant in token space whose information content is known exactly, is large enough to be detectable under the pre-registered branch condition (0.7–1.1 nats), and does not perturb the carrier's marginal distribution enough to be detectable by that route instead.

A first design, retired. The natural construction — a deterministic translation of one digit by an amount determined by another — cannot carry enough information on this carrier. The carrier's coarse digit marginals are already near-uniform: measured entropies of 2.3938, 2.1796 and 2.1873 nats against maxima of $\ln 11 = 2.3979$ and $\ln 9 = 2.1972$, i.e. 99.2–99.8% of maximum. Under near-uniform marginals a translation plant is capped at 0.2769 nats, below the required band, so it was retired as information-poor by construction rather than tuned.

The adopted design. The target digit is redrawn from a discretized Gaussian on the level grid, centred monotonically on the source level, with width σ_{plant} solved so that the exact mutual information lands in band. Table 3 gives the calibration receipt at solution and the same quantities recomputed over the full training stream. The closed-form and Monte-Carlo estimates agree to 9×10^{-4} at calibration and to 2×10^{-5} over the full stream; the marginal KL between the planted and carrier target distributions is 0.005 and the entropy shift is -0.005 nats, so the plant is not detectable from the marginal alone; and an unplanted control run returns a rank correlation of -8.8×10^{-5}, confirming the probe reads zero when there is nothing to read.

Table 3: Token-space plant, calibration and full-stream receipts. The plant is solved to a target information content rather than tuned to a result; the oracle ceiling is the rank correlation an optimal predictor of the planted rule would achieve, and bounds what any model can reach.

quantity	at calibration	full stream
plant width σ_{plant}	1.0077	—
mutual information, closed form (nats)	0.9000	0.9003
mutual information, Monte-Carlo plug-in (nats)	0.9009	0.9003
$\lvert$closed form − Monte-Carlo$\rvert$	9.1×10^{-4}	2.0×10^{-5}
marginal KL, planted vs. carrier	0.00476	0.00480
entropy shift ΔH (nats)	−0.0054	−0.0052
oracle rank-correlation ceiling	0.926	—
unplanted control, rank correlation	-8.8×10^{-5}	—

C Two defects in the legibility gate

Both defects are instances of the same failure — a check that cannot fail is not a check — and both were closed before the gate was relied upon. They are recorded because the gate is a hard stop on expenditure, and because the second one was material.

A skipped dimension could not lower the verdict. A dimension with too few unmasked bars to measure took a code path that continued the loop *before* the pass flag was accumulated. A gate run in which all six microstructure dimensions were too thin therefore returned `pass: True` having measured nothing, and a run with five thin and one passing dimension returned `pass: True` on a gate whose stated contract is all six. Both were reproduced before the fix. "We could not measure the channel this gate exists to measure" is now a third outcome, and it halts.

The probe read one symbol of two hundred. The probe took a fixed 150,000-row head of a symbol-ordered concatenation with a contiguous train/validation cut. On the production data that window spans *one of* 200 *symbols*, so the check governing all second-stage expenditure was being made on a single instrument. The naive repair — shuffling before the split — would have broken the other property that matters, since one-minute bars are autocorrelated and an interleaved split puts near-duplicate neighbours on both sides. The fix is a per-symbol stratified split that is blocked in time *within* each symbol, which preserves both coverage and blocking.

The materiality of the two legs differs, and we measured rather than asserted it. On the production data the minimum per-dimension unmasked bar count across all 200 symbols is 277,758 against the gate's 10,000-bar floor, with zero symbols below it, so the all-thin case cannot arise here: the first defect was real but immaterial on this data. The second was severe.

D Reproducibility and verification

[pinned decision surface; content hashes; environment lockfiles; amendment log.]

E Claims audit

[measured-versus-estimated table.]